

## VARSITY CYBER CLASH NETWORK DESIGN AND IMPLEMENTATION REPORT #

### Executive Summary 1.0 ##

*The Varsity Cyber Clash project aims to design a secure, scalable, and high-performance network infrastructure capable of supporting a large-scale university E-Sports tournament.*

*The network must simultaneously accommodate competitive gamers, tournament administrators, broadcasters, spectators, and critical server services while ensuring low .latency, reliability, and efficient resource utilization*

*To satisfy these requirements, a Three-Tier Hierarchical Star Topology has been implemented together with VLAN segmentation, Variable Length Subnet Masking (VLSM), inter-VLAN routing, Quality of Service (QoS), centralized server services, and network monitoring. The proposed design provides strong security, simplified management, efficient .IP address allocation, and future scalability*

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### Project Requirements Analysis 2.0 #

*The tournament environment consists of several user groups with different networking :requirements*

*.Spectators: 300 wireless users requiring Internet connectivity and live score updates •*

*.Gamers: 70 players requiring low-latency and high-priority network access •*

*Administrators and Broadcasters: 30 users requiring access to tournament management • .resources*

*Server Farm: Critical infrastructure hosting game servers, authentication services, • .streaming platforms, and tournament systems*

*Management Team: Network administrators responsible for monitoring and maintaining • .network operations*

*These requirements create significant differences in traffic patterns, security levels, and bandwidth demands. Therefore, network segmentation and efficient addressing are essential .components of the design*

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### Network Topology Design 3.0 #

#### Topology Selection 3.1 ##

*:The network adopts a Three-Tier Hierarchical Star Topology consisting of*

Core Layer .1  
Distribution Layer .2  
Access Layer .3

:This topology was selected because it provides

- High scalability •
- Simplified troubleshooting •
- Improved fault isolation •
- Better traffic management •
- Easier future expansion •

Compared with a flat network architecture, the hierarchical design significantly improves  
.performance and manageability for large-scale deployments

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### Core Layer 3.2 ##

.The Core Layer acts as the backbone of the network

:Components

- (ISP Fibre Internet Connection (1 Gbps •
- Core Router •

:The Core Router performs several critical functions

- Internet Gateway •
- (Network Address Translation (NAT •
- (Access Control Lists (ACLs •
- Firewall Filtering •
- Inter-VLAN Routing •

The router ensures secure communication between internal VLANs while providing  
.controlled access to the Internet

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### Distribution Layer 3.3 ##

.The Distribution Layer consists of a Layer 3 Managed Switch

:Responsibilities include

- VLAN Trunking •
- (Dynamic Routing (OSPF/RIP •
- QoS Implementation •

- Traffic Aggregation •*
- Routing Between Network Segments •*

*The Layer 3 switch acts as the central traffic control point within the network and provides efficient routing services between departments*

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#### *Access Layer 3.4 ##*

*.The Access Layer provides direct connectivity for users and devices*

*:Dedicated access switches are deployed for*

- Players Network •*
- Server Farm •*
- Administrators and Broadcasters •*
- Wireless Spectator Network •*
- Management Network •*

*.This arrangement reduces congestion and improves traffic organization*

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#### *VLAN Segmentation Strategy 4.0 #*

##### *Purpose of VLAN Segmentation 4.1 ##*

*Virtual Local Area Networks (VLANs) are used to logically separate departments operating on the same physical infrastructure*

*:Advantages include*

- Enhanced security •*
- Reduced broadcast traffic •*
- Improved performance •*
- Simplified management •*
- Better fault isolation •*

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##### *VLAN 10 – Players Network 4.2 ##*

*:Purpose*

*.Provides connectivity for competitive gaming devices and broadcast rigs*

*:Characteristics*

- High-performance wired network •*
- QoS priority enabled •*
- Low-latency communication •*

*:Devices*

- Gaming PCs 70 •*
- Broadcast and Streaming Rigs •*

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#### *VLAN 20 – Admin and Broadcaster Network 4.3 ##*

*:Purpose*

*.Supports tournament administration and event broadcasting activities*

*:Devices*

- Admin Workstations •*
- Broadcaster Stations •*
- Network Printer •*

*.Security controls ensure only authorized personnel can access this VLAN*

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#### *VLAN 30 – Server Farm Network 4.4 ##*

*:Purpose*

*.Hosts mission-critical tournament infrastructure*

*:Benefits*

- Centralized service delivery •*
- Easier maintenance •*
- Enhanced security •*
- Reduced management complexity •*

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#### *VLAN 40 – Spectator Wireless Network 4.5 ##*

*:Purpose*

*.Provides wireless connectivity for spectators*

*:Characteristics*

- Supports approximately 300 devices •*
- Wi-Fi 6 infrastructure •*
- Guest access environment •*

*.This VLAN receives the largest subnet allocation due to the high number of users*

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*VLAN 99 – Management Network 4.6 ##*

*:Purpose*

*.Dedicated network for infrastructure administration*

*:Devices*

- Network Administration Terminal •*
- Monitoring Systems •*

*.Only authorized administrators can access this VLAN*

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*(IP Addressing Design (VLSM 5.0 #*

*Justification for VLSM 5.1 ##*

*Variable Length Subnet Masking (VLSM) was selected because different departments  
.require different numbers of hosts*

*.Using identical subnet sizes would waste address space*

*:VLSM provides*

- Efficient address allocation •*
- Reduced wastage •*
- Better scalability •*
- Professional network design •*

*The private Class A address block 10.0.0.0/8 was selected because it provides a large  
.address space suitable for future expansion*

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*Host Requirement Analysis 5.2 ##*

*Spectators = 300 Hosts*

*Gamers = 70 Hosts*

*Admins = 30 Hosts*

*Server Farm = 20 Hosts*

*Management = 5 Hosts*

*.Subnets were allocated from largest to smallest to maximize efficiency*

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*VLAN Addressing Plan 5.3 ##*

*VLAN 40 – Spectators ###*

*Network Address: 10.0.0.0/23*

*Subnet Mask: 255.255.254.0*

*Default Gateway: 10.0.0.1*

*Usable Hosts: 510*

*Host Range: 10.0.0.2 – 10.0.1.254*

*Broadcast Address: 10.0.1.255*

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*VLAN 10 – Players ###*

*Network Address: 10.0.2.0/25*

*Subnet Mask: 255.255.255.128*

*Default Gateway: 10.0.2.1*

*Usable Hosts: 126*

*Host Range: 10.0.2.2 – 10.0.2.126*

*Broadcast Address: 10.0.2.127*

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*VLAN 20 – Admin and Broadcasters ###*

*Network Address: 10.0.2.128/27*

*Subnet Mask: 255.255.255.224*

*Default Gateway: 10.0.2.129*

*Usable Hosts: 30*

*Host Range: 10.0.2.130 – 10.0.2.158*

*Broadcast Address: 10.0.2.159*

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*VLAN 30 – Server Farm ###*

*Network Address: 10.0.3.0/27*

*Subnet Mask: 255.255.255.224*

*Default Gateway: 10.0.3.1*

*Usable Hosts: 30*

*Host Range: 10.0.3.2 – 10.0.3.30*

*Broadcast Address: 10.0.3.31*

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*VLAN 99 – Management ###*

*Network Address: 10.0.99.0/29*

*Subnet Mask: 255.255.255.248*

*Default Gateway: 10.0.99.1*

*Usable Hosts: 6*

*Host Range: 10.0.99.2 – 10.0.99.6*

*Broadcast Address: 10.0.99.7*

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## *Server Farm Infrastructure 6.0 #*

*.The Server Farm contains all essential tournament services*

### *Game Server ##*

*.Provides dedicated match hosting and low-latency gameplay*

### *Web and Registration Server ##*

*.Manages participant registration and event information*

### *FTP Server ##*

*.Provides file distribution and resource sharing*

### *Live Streaming Server ##*

*.Supports real-time HD broadcasting*

### *Tournament Management System ##*

*.Manages tournament brackets and score tracking*

### *VoIP Server ##*

*.Provides secure voice communication between players*

### *Caching Server ##*

*.Reduces bandwidth consumption and improves performance*

### *Active Directory Server ##*

*.Provides centralized authentication and access control*

### *Live Scoreboard Server ##*

*.Delivers real-time tournament updates to spectators*

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## *Hardware Infrastructure Justification 7.0 #*

### *Core Router ##*

*.The router provides Internet connectivity, NAT, firewall services, and inter-VLAN routing*



### *Layer 3 Managed Switch ##*

*.Supports VLAN routing, QoS, dynamic routing, and traffic aggregation*

### *Access Switches ##*

*.Provide reliable Gigabit connectivity for users and servers*

### *Wireless Access Points ##*

*.Wi-Fi 6 access points provide high-density wireless coverage for spectators and staff*

### *Monitoring Dashboard ##*

*.Provides real-time network visibility using SNMP monitoring and fault detection tools*

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### *Security Design 8.0 #*

*.Security is a critical requirement due to the large number of participants and public users*

*:Security mechanisms include*

- VLAN Segmentation •*
- Access Control Lists •*
- Firewall Protection •*
- NAT •*
- Active Directory Authentication •*
- Management VLAN Isolation •*
- Secure Administrative Access •*
- Network Monitoring •*

*.These controls reduce unauthorized access and improve overall network integrity*

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### *Network Traffic Flow 9.0 #*

#### *Player Accessing Game Server ###*

*Gaming PC → Access Switch → Layer 3 Switch → Server Farm → Game Server*

#### *Spectator Accessing Internet ###*

*Mobile Device → Access Point → Wireless Controller → Router → ISP*

### *Administrator Accessing Tournament System ###*

*Admin Workstation → Access Switch → Layer 3 Switch → Tournament Management Server*

*These paths demonstrate efficient communication while maintaining security boundaries  
.between VLANs*

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### *Future Expansion 10.0 #*

*The design intentionally utilizes only a small portion of the available 10.0.0.0/8 address  
.space*

*:Future enhancements may include*

- Additional gaming halls •*
- Larger spectator capacity •*
- Additional tournament servers •*
- Redundant Internet connections •*
- Secondary data center deployment •*
- Cloud integration •*

*.This ensures long-term scalability and investment protection*

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### *Conclusion 11.0 #*

*The Varsity Cyber Clash network successfully fulfills all functional, performance, security,  
and scalability requirements of a modern E-Sports tournament environment. Through the  
implementation of a Three-Tier Hierarchical Star Topology, VLAN segmentation, VLSM  
addressing, centralized server infrastructure, QoS optimization, and comprehensive security  
controls, the proposed solution delivers reliable and efficient network services to all  
.stakeholders*

*The design demonstrates professional networking practices, efficient resource utilization,  
and readiness for future expansion, making it a robust and enterprise-grade solution for  
.large-scale competitive gaming events*